Mandatory project in Digital Signal Processing

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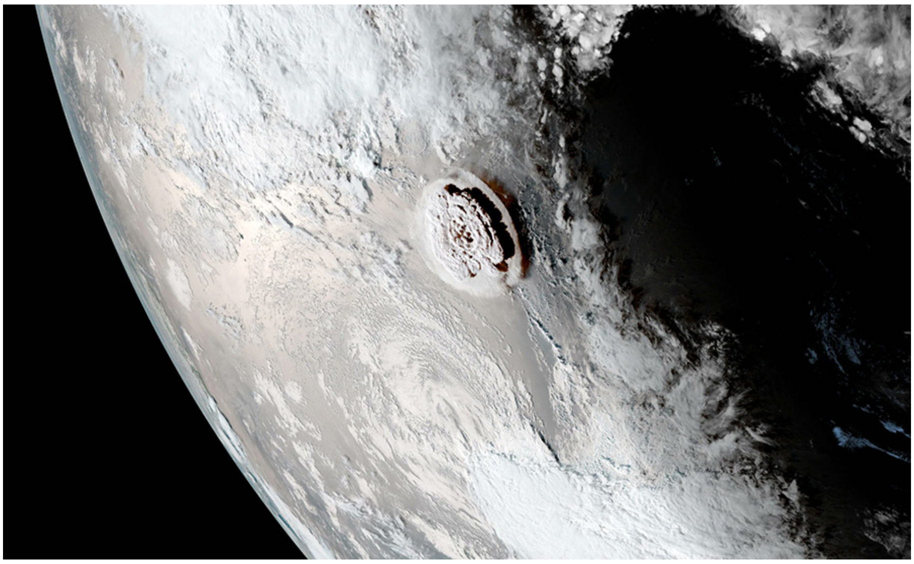

The volcanic eruption, captured by NOAA's GOES-West satellite.
Credit: GOES-West NOAA/RAMMB/CIRA. Figure from Grant (2022).

The volcanic eruption on 15 January 2022 eruption of the Hunga, Tonga–Hunga Haʻapai, hereafter referred as Hunga Tonga, produced an extraordinary blast that was probably the most powerful in the last century (Vergoz *et al.*, 2022).

Atmospheric waves generated from this event were of such remarkable magnitude that they were seen to propagate four times around the Earth over six days (Matoza *et al.*, 2022).

This remarkable event was so immense that, under optimum propagation and noise conditions, effects from it it could be heard by human ears as far away as in Alaska.

In the current project, you will work on real sensor recordings related to Hunga Tonga. If you succeed with the given tasks, the results of your analysis will be in accordance with findings in the *Science* paper Matoza *et al.* (2022).

This extraordinary event and the unique set of signals it produced continues to spark new geophysical discoveries and research. These propagating waves have allowed researchers to gain more knowledge about, for example, Earth's atmosphere, tsunami generation and propagation and volcanic plume dynamics. As an illustration, we note that the number of research papers indexed by Google Scholar that mention "Hunga Tonga" in 2022 and 2025 is greater than 3100 [See the following URL: `https://scholar.google.com/scholar?as_ylo=2022&q=%22hunga+tonga%22&hl=en&as_sdt=0,5`].
A nice 10-minute video about the event is available on YouTube: `https://www.youtube.com/watch?v=sZZVVwqZ0rs`.

Contents

1 The Hunga Tonga volcanic event

It is highly recommended that you read the Abstract and Introduction parts of the three papers Vergoz *et al.* (2022); Matoza *et al.* (2022); Wright *et al.* (2022) – direct access links are available from the Bibliography in the end of the current project description. This way, you will get a basic understanding of various global waves and physics induced by this extraordinary volcanic event. You will also get exposed to some of the various probing technologies that help us observe atmospheric waves, such as those that this event generated.

2 Background info on the dataset and how to download it

This project takes advantage of work done by Eckel *et al.* (2022) that was presented at an international scientific meeting in 2022, the *European Geosciences Union General Assembly*. The Python codes used for the Eckel *et al.* (2022) presentation are openly available at `https://github.com/felix-eckel/infrasection`, although it will not be required that you utilize these directly.

In particular, we are interested in analyzing time series from a set of sensors that measure the atmospheric pressure perturbation occurring when these massive atmospheric waves from the volcano pass over the ground-based stations that host the sensors.

- The repository `https://github.com/felix-eckel/infrasection` provides a data downloader tool for downloading such data from openly available international data centers, for example, `https://www.iris.edu/hq/`.

  Note, however, that you will not need to run the code from this github repository in order to solve the current project tasks.

- Figure 1 displays a selected subset of 201 stations where the data you will analyze were recorded.

- In order to avoid bloating the data centers with requests from all students of our course, all data has already been downloaded and made available from the following course URL at the University of Oslo servers: `https://www-int.uio.no/studier/emner/matnat/ifi/IN3190/h25/login-required/project/data/project_data.zip`.

  - To facilitate reading these data using different platforms and coding languages (Matlab / Python / Julia / R / etc.) that the students might prefer to use, we have serialized the original miniSEED format files into the universal HDF5 file format. This was achieved using the ObsPyH5 package (`https://github.com/trichter/obspyh5`).

    Preparation task: Download the HDF5 files with sensor time series from this link: `https://www-int.uio.no/studier/emner/matnat/ifi/IN3190/h25/login-required/project/data/project_data.zip` (around 1.1 GB size)

  - In addition to a zip archive with all sensor time series, the same `data/` folder [`https://www-int.uio.no/studier/emner/matnat/ifi/IN3190/h25/login-required/project/data/`] includes the Matlab scripts `set_groot.m` and `run_provided.m`.
    Preparation task: Download also these scripts.

  - You can use `set_groot.m` (or a modification thereof) to adjust your plot setting defaults. `run_provided.m` can serve as a starting point for your data processing.

    Our previous student Markus Bjørklund has kindly shared some Python code snippets that can help you translate the `run_provided.m` code into Python if you choose to code in this language instead. See `bjorklund_extracts.py`.

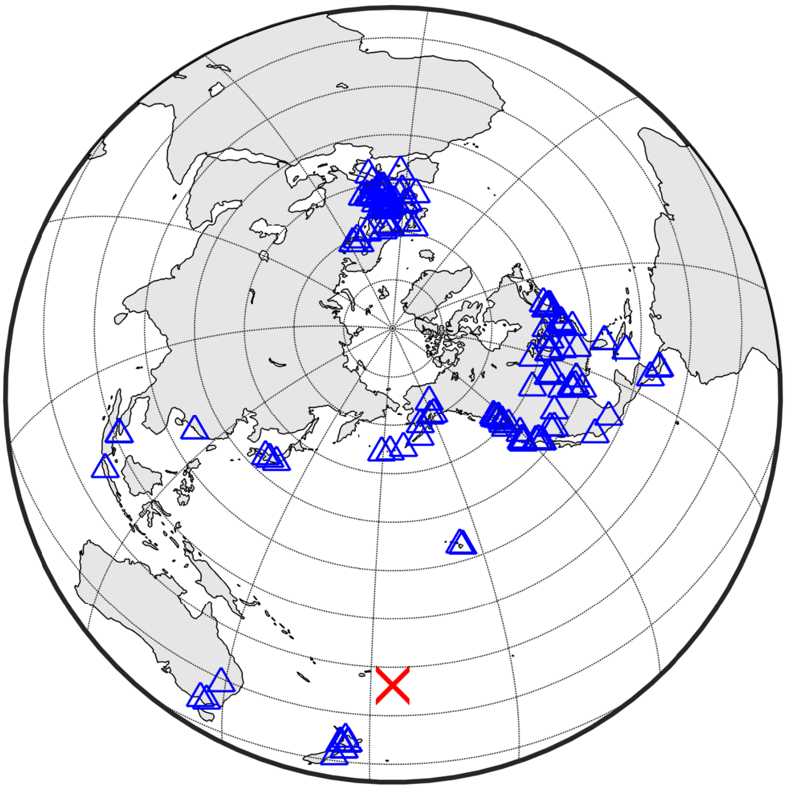

Figure 1: Geographic locations of the stations where the data you'll process were recorded (blue triangles), as well as the location of Hunga Tonga (red cross).

3 Your mission

3.1 Submission formatting requirements and submission contents

Please take the following conditions into account when you submit your solutions to this project:

- All code must be properly commented so we easily can understand your approaches.
- All code must be attached in your submission.
- You should set up a "run script" that, without input parameters, generates all figures and output needed. This can also be done in form of a Live script or Jupyter notebook, or it can be a standalone .m or .py script file. It shall be immediately discernable from the code commenting where in the programs that solutions to each task/subtask are implemented.
- All plots must have large enough fontsizes and line widths. Normally, all text in the figures shall be at least the size of the main text body font.
- The report shall clearly indicate what tasks you are solving.
- The report shall include figures and appropriate links to the code.
- Figures shall be saved in vectorized formats, e.g., PDF or SVG. If you use Matlab, it is advisable to save figures using the `export_fig` package or `exportgraphics` directly into vectorized output (typically in PDF), instead of other export approaches.
- The report shall be typeset (and not handwritten), preferably using LaTeX, Markdown, or similar markup-based frameworks. Another alternative can be that the report is fully integrated in a submitted Live script or Jupyter notebook.
- Don't hesitate to use the *Discourse* forum to ask for advice if you're about to get stuck.
- If you have not yet programmed in Matlab, it can be worth using this opportunity gain some experience in this language. But we also read Python and Julia, for example.

3.2 The actual Project tasks

Task 1 — The overview

a) Run the script `run_provided.m` (or an adaption of it to your favorite coding language) in order to parse the dataset and make some basic processing and visualization. Make sure you understand all components of the script. Include the map plot in your report. The HDF data format includes several metadata information fields where, for example, the latitude and longitude locations of each station is provided.

b) What are the shortest and longest *great circle* distances between Hunga Tonga and the stations?

c) Provide a plot with a display of all (sorted) distances between the event location and the station locations.

Task 2 — Data filtering

a) The `run_provided.m` script defines three different FIR filter impulse responses: $h_1[n]$, $h_2[n]$, $h_3[n]$. Plot these.

b) When filtering an input signal $x[n]$ using an FIR filter with impulse response $h[n]$, the output $y[n]$ can be found using a convolution operation:

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]. \tag{1}$$

Code up a function or subroutine which performs this convolution given the input variables `h` and `x`, as well as an additional parameter `ylen_choice`. If `ylen_choice == 1`, the output of the convolution shall be of length $M+N-1$, where M and N are the lengths of $x[n]$ and $h[n]$, respectively. For `ylen_choice == 0`, the output shall be of the same length as $x[n]$.

This means that for `ylen_choice == 0`, it would reproduce the result of the Matlab `conv` function with the `shape` parameter being set to `'same'`, see the documentation here `https://se.mathworks.com/help/matlab/ref/conv.html#bucr8kb-1-shape`, and you can of course use this function directly when you verify your own implementation.

Verify that your code works as intended by showing a toy signal example with direct comparison to output from already existing convolution code from Matlab / SciPy / DSP.jl, for example.

c) The frequency spectrum, or discrete-time Fourier transform (DTFT), $H(e^{j\omega})$ of a signal or sequence $h[n]$ is defined as

$$H(e^{j\omega}) = \sum_{n=-\infty}^{\infty} h[n]e^{-j\omega n}. \tag{2}$$

Code up a function or subroutine which performs this operation given the input signal `h`.

In addition, the function shall take as input `N` and `fs`, which are the number of points on the unit circle that are output for ω and the sampling frequency, respectively.

The function shall provide two outputs: the complex vector $H(e^{j\omega})$, as well as the frequency axis vector provided in physical frequency and the unit Hertz.

d) Display the absolute value of the frequency spectrum of the provided impulse responses $h_1[n]$, $h_2[n]$, $h_3[n]$. That is, you shall plot $|H_1(e^{j\omega n})|$, $|H_2(e^{j\omega n})|$, and $|H_3(e^{j\omega n})|$ as function of the physical frequency F.

e) Which of $h_1[n]$, $h_2[n]$, $h_3[n]$ is lowpass, bandpass, or highpass filters? [See, for example, page 222–224 in Manolakis & Ingle for illustrations of (idealized) lowpass, bandpass, and highpass filters.]

f) Filter all 201 input time series using filters $h_1[n]$, $h_2[n]$, $h_3[n]$ and store the respective outputs for each operation. Although you have previously coded up your own convolution function, to get a faster execution of your filtering, it can be advisable that you perform the filtering using builtin convolution code (in Matlab / SciPy / DSP.jl, for example).

   You are fully free to reorganize the infrastructure and data holding approach selected in `run_provided.m`. For example, if you dislike putting the data vectors into a struct, you can choose another approach.

g) Now we'll generate results similar to what is shown in Figure 2C of the *Science* paper Matoza *et al.* (2022) (reproduced in Figure 2 of the current document).

   Figure 2C of Matoza *et al.* (2022) shows data bandpass filtered between 0.01 and 2 Hz. Choose one of the filters $h_1[n]$, $h_2[n]$, $h_3[n]$ which you find the most suitable, and provide the reasons why you made this choice.

   Generate a *section plot* in style with Figure 2C of Matoza *et al.* (2022). If you need inspiration on how this can be coded up, you can take a look at the way this kind of plotting is implemented in `https://docs.obspy.org/_modules/obspy/imaging/waveform.html#WaveformPlotting.plot_section`.

   You shall have time on the x axis and distance between Hunga Tonga and respective station on the y axis.

Task 3 — Estimating the celerity of the globally propagating acoustic wave

For a wave that propagates a (prospectively complex) path in the atmosphere, we define *celerity* as the great circle horizontal distance covered, divided by the time spent. This typically has the unit m/s.

a) Write a routine which loops over all station signals, sorted by distance from Hunga Tonga. It can also be useful that the routine takes two additional inputs which tell at which station index to start and which station index to stop the analysis.

   The loop shall do the following:

   - Open a new figure window
   - Plot four subplots: the raw trace, the trace filtered by $h_1[n]$, the trace filtered by $h_2[n]$, and the trace filtered by $h_3[n]$.
   - Activate a pointer, for example using the `ginput` function. Now you shall manually click on, or *pick*, what appears to be the arrival time of the higher-frequency infrasound wave on this station. The arrival time estimate shall be stored a vector for later usage. This process is called *arrival time picking*.
   - Close the figure window. The vector of picked arrival times shall be saved in a file which can later be read and further analyzed.

     As you get used to the process of picking arrival times, you might later want to re-run your picking for a subset of of the stations.

     It is therefore advisable that each "picking session" you run saves the output to its own file, and that for each loop iteration, the vectors of picked arrival times is saved along with the vector of station indices processed.

b) Now write a plotting and analysis routine that uses the saved "arrival time picks" for display and for further calculation:

- This shall set up a figure and plot the estimated travel times on the x axis and the event-to-station distance on the y axis.
- Some station data are too noisy, or might include a competing arrival from some other source that is dominating over the Hunga Tonga signals. Display the picks that appear as obvious outliers in a contrasting color and symbol, and disregard these outliers in your further analysis.
- Fit the time versus distance data to a straight line. The slope of this fitted line is your *celerity* estimate. If the distance is provided in meters and the time in seconds to your straight line fitting routine, the celerity hence comes in meters per second.

  How well does your estimate compare to typical celerities of infrasound waves provided on page 3 of Vergoz *et al.* (2022) or page 3 of Matoza *et al.* (2022)?

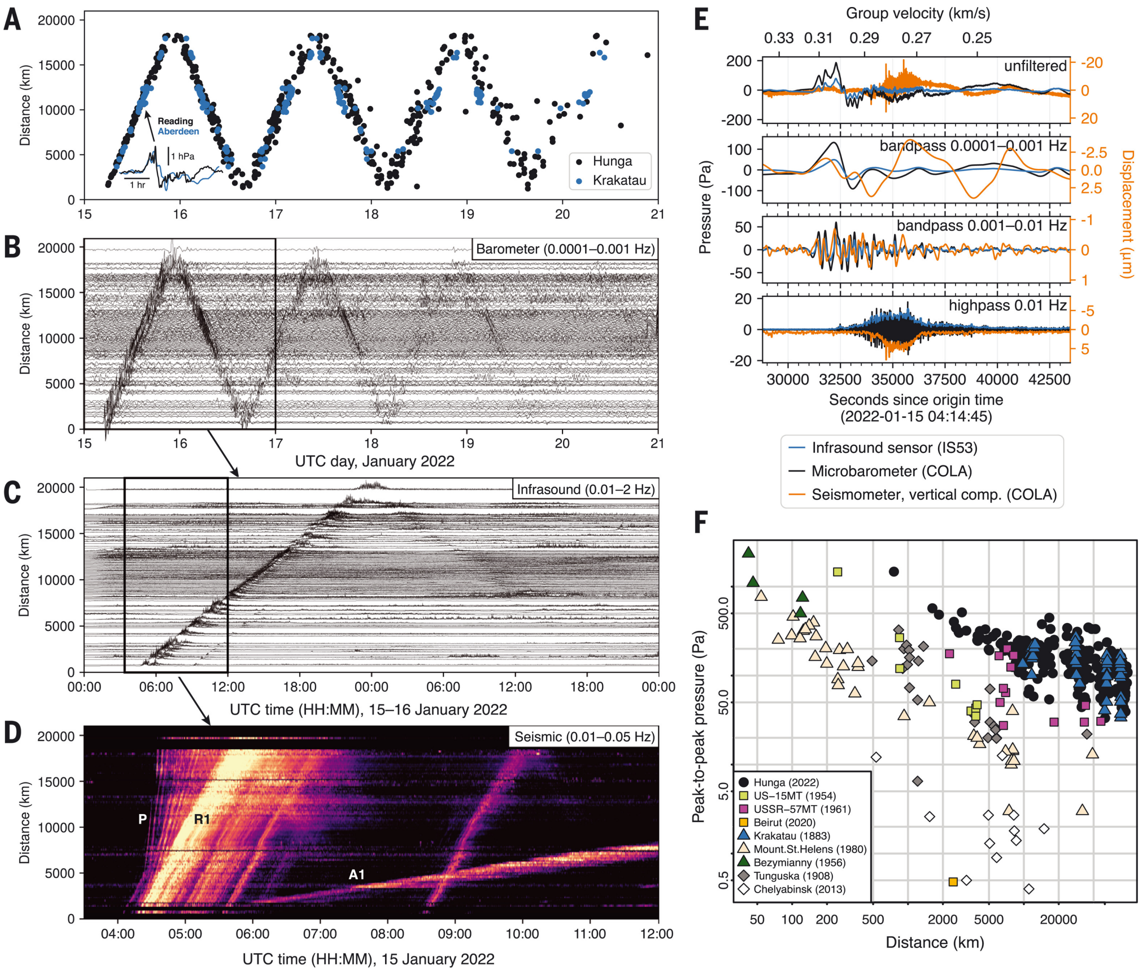

Figure 2: Ground-based observations. (A) Lamb wave arrival times for 2022 Hunga (black) compared to 1883 Krakatau (blue); inset: Lamb A1 arrival waveform comparison (3). Global record sections of (B) barometer, (C) infrasound, and (D) seismic data showing the multiple arrivals and wave passages (see Fig. 1A inset); waveforms aggregated by radial distance (Fig. S7). A separate Rayleigh R1 is associated with the later 08:31 event. (E) Colocated microbarometer (black), infrasound sensor (blue), and seismometer (orange) waveforms; lower panel shows inverted displacement envelope. (F) Wideband peak-to-peak pressure versus distance comparing 2022 Hunga with large historical explosive events. (Reproduced from Matoza *et al.* (2022).)

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